CSCI 651 Homework 2: Ping and Traceroute

1. Ping
    a. When an address is pinged, it will continuously ping the address until a keyboard interrupt is performed

```
PS C:\Users\owner\Documents\School Assignments\Masters\Fall2026\CSCI51_FoundOfCompNetworks\Labs\Computer_Networks_Homework_Two> python my_ping.py 8.8.8.8
Reply from 8.8.8.8: seq=1 time=29.46ms
Reply from 8.8.8.8: seq=2 time=23.48ms
Reply from 8.8.8.8: seq=3 time=121.35ms
Reply from 8.8.8.8: seq=4 time=54.52ms
Reply from 8.8.8.8: seq=5 time=17.20ms
Reply from 8.8.8.8: seq=6 time=54.20ms
Reply from 8.8.8.8: seq=7 time=39.92ms
Reply from 8.8.8.8: seq=8 time=37.74ms
```

    b. When an address is pinged with the -c flag, it will only ping until the designated number of packets is received

```
PS C:\Users\owner\Documents\School Assignments\Masters\Fall2026\CSCI51_FoundOfCompNetworks\Labs\Computer_Networks_Homework_Two> python my_ping.py -c 5 8.8.8.8
Reply from 8.8.8.8: seq=1 time=22.02ms
Reply from 8.8.8.8: seq=2 time=19.02ms
Reply from 8.8.8.8: seq=3 time=99.05ms
Reply from 8.8.8.8: seq=4 time=22.30ms
Reply from 8.8.8.8: seq=5 time=69.12ms
```

    c. When an address is pinged with the -t flag, it will continuously ping the address, until it is timed out at the designated time

```
PS C:\Users\owner\Documents\School Assignments\Masters\Fall2026\CSCI51_FoundOfCompNetworks\Labs\Computer_Networks_Homework_Two> python my_ping.py -t 2.0 8.8.8.8
Reply from 8.8.8.8: seq=1 time=109.22ms
Reply from 8.8.8.8: seq=2 time=40.59ms
```

    d. An address can be pinged with a designated packet size, designated by the -s flag

```
PS C:\Users\owner\Documents\School Assignments\Masters\Fall2026\CSCI51_FoundOfCompNetworks\Labs\Computer_Networks_Homework_Two> python my_ping.py -s 82 8.8.8.8
Reply from 8.8.8.8: seq=1 time=29.89ms
Reply from 8.8.8.8: seq=2 time=41.30ms
Reply from 8.8.8.8: seq=3 time=19.62ms
Reply from 8.8.8.8: seq=4 time=17.86ms
Reply from 8.8.8.8: seq=5 time=14.88ms
Reply from 8.8.8.8: seq=6 time=33.80ms
Reply from 8.8.8.8: seq=7 time=22.80ms
Reply from 8.8.8.8: seq=8 time=18.41ms
Reply from 8.8.8.8: seq=9 time=18.61ms
Reply from 8.8.8.8: seq=10 time=23.44ms
Reply from 8.8.8.8: seq=11 time=17.19ms

 End of Ping
```

2. Traceroute (I was only receiving timeouts)
    a. When traceroute is called, it sends out a UDP packet probe

```
PS C:\Users\owner\Documents\School Assignments\Masters\Fall2026\CSCI51_FoundOfCompNetworks\Labs\Computer_Networks_Homework_Two> py my_traceroute.py 8.8.8.8
1 * * *
2 * * *
3 * * *
4 * * *
5 * * *
6 * * *
7 * * *
8 * * *
9 * * *
10 * * *
11 * * *
12 * * *
13 * * *
14 * * *
15 * * *
```

b. When -S is ran, its prints a summary of how many probes were not answered

```
PS C:\Users\owner\Documents\School Assignments\Masters\Fall2026\CSCI51_FoundOfCompNetworks\Labs\Computer_Networks_Homework_Two> py my_traceroute.py -S 8.8.8.8
1 * * *
3 probes unanswered
2 * * *
3 probes unanswered
3 * * *
3 probes unanswered
4 * * *
3 probes unanswered
```

c. When ran with the -q flag is raised, it changes how many probes are sent for each hop

```
PS C:\Users\owner\Documents\School Assignments\Masters\Fall2026\CSCI51_FoundOfCompNetworks\Labs\Computer_Networks_Homework_Two> py my_traceroute.py -S -q 5 8.8.
1 * * * * *
5 probes unanswered
2 * * * * *
5 probes unanswered
```